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HARRISDALE
Senior High School

Harrisdale Senior High School

Year 10 EXTENSION Mathematics

Course Outline 2026

Term 1 – 4

Text Required: Cambridge * Essential Mathematics 10/10A for the Australian Curriculum

Supplementary Resource: Mathspace Year 10/10A Australian Curriculum

THIS OUTLINE IS SUBJECT TO CHANGE



Year 10 Achievement Standard

Year 10 Achievement Standard

By the end of the year:

Number and algebra

Students compare the difference in using approximate to exact values on final calculations. They substitute values into real-life formulas to find unknowns using digital tools. Students solve one variable linear inequalities, verify the solution and represent on a number line. They solve linear simultaneous equations graphically. Students graph quadratic and exponential functions and relate key graphical and algebraic features. They calculate income tax and use repeated simple interest to determine compound interest.

Measurement and geometry

Students use Pythagoras' theorem and/or trigonometry to determine unknown sides and angles in right-angled triangles involving angles of elevation and depression. They determine the effect on perimeter, area and volume when shapes and objects are enlarged or reduced. They use triangle and angle properties to show reasoning as to why triangles are similar and find unknown sides and angles in similar triangles. They determine volume, capacity and surface area of composite objects.

Probability and statistics

Students choose and construct appropriate sample spaces for two- and three-stage chance experiments and assign probabilities to events involving conditional statements. They conduct chance experiments and simulations to model conditional probability. They describe possible association between variables represented in a scatterplot or two-way table. Students represent multiple datasets using boxplots and compare shape, spread and centre. They critically analyse claims, inferences and conclusions in the media and identify potential sources of bias.

TERM ONE			
Week	Content	Cambridge ESSENTIAL FOR WA	Assessment
LINEAR FUNCTIONS			
1	<u>Content Descriptors</u> Move flexibly between real number inequalities expressed as a worded statement, algebraically or on a number line Solve one-variable linear inequalities involving brackets and/or a variable on each side. Represent the solution on a number line and verify the solution by substitution <u>Success Criteria</u> Linear Inequalities • Translate flexibly between real-life worded statements (e.g., "maximum speed of 80"), algebraic inequalities (e.g., $x < 80$), and number line representations. • Solve one-variable linear inequalities that involve brackets and/or variables on both sides of the inequality symbol. • Represent solutions on a number line using inequalities ($<$, $>$) and ($\leq$, $\geq$). • Verify solutions by substituting values back into the original inequality.	Chapter 1A – 1C	

2	<u>Content Descriptors</u> Function notation Determine the solution to linear simultaneous equations in the forms $y = mx + c$ or $ax + by = c$ graphically and verify the solution by substitution Determine the solution to linear simultaneous equations in the forms $y = mx + c$ or $ax + by = c$ algebraically and verify the solution by substitution or using digital tools <u>Success Criteria</u> Simultaneous Equations - Determine the solution to linear simultaneous equations graphically by plotting lines and identifying the coordinates of the point of intersection. - Solve linear simultaneous equations algebraically using the Substitution Method and the Elimination Method (for equations in forms $y = mx + c$ or $ax + by = c$). - Verify solutions by substitution or using digital tools.	Chapter 1D – 1F, 1J, 1K	
3	<u>Content Descriptors (continued)</u> Function notation Determine the solution to linear simultaneous equations in the forms $y = mx + c$ or $ax + by = c$ graphically and verify the solution by substitution Determine the solution to linear simultaneous equations in the forms $y = mx + c$ or $ax + by = c$ algebraically and verify the solution by substitution or using digital tools	Chapter 1J, 1K, 1L	

	<u>Success Criteria (continued)</u> Simultaneous Equations - Determine the solution to linear simultaneous equations graphically by plotting lines and identifying the coordinates of the point of intersection. - Solve linear simultaneous equations algebraically using the Substitution Method and the Elimination Method (for equations in forms $y = mx + c$ or $ax + by = c$). Verify solutions by substitution or using digital tools.		
4	<u>Content Descriptors</u> Use gradient and/or point/s to graphically and algebraically determine equations of parallel and perpendicular lines Identify and distinguish between linear, quadratic and exponential functions represented by equations, tables of values and graphs Modelling with number and algebra In real-world situations involving real numbers, absolute and percentage error, linear inequalities, simultaneous equations, real-world formulas, quadratic or exponential functions, taxation and/or compound interest - analyse the situation, decide if an exact or approximate solution is required and determine assumptions and constraints - represent the situation mathematically in order to reach a solution - interpret and communicate findings in terms of the context and any assumptions or constraints	Chapter 1H, 1I	

	<u>Success Criteria</u> Linear Relationships and Graphs - Identify and distinguish between linear, quadratic, and exponential functions by analysing representations in equations, tables of values, and graphs. - Use gradients and/or points to graphically and algebraically determine the equations of parallel and perpendicular lines. (EXTENSION)		
5	<u>Content Descriptors</u> Review contents from Weeks 1 – 5 Assessment 1 – Linear Functions (TEST)		Week 5 Assessment 1 – Linear Functions (TEST) Returned by Week 7
UNIVARIATE AND BIVARIATE DATA			
6	<u>Content Descriptors</u> Represent and analyse boxplots. Explain differences between multiple boxplot datasets in terms of shape, spread and centre. Compare or match the shapes of boxplots to distributions depicting the same data	9A – 9E	

	Determine the mean and standard deviation of a dataset. Investigate, analyse and interpret the effect of individual data values, including outliers, on the standard deviation Critically analyze the claims, inferences and conclusions of statistical reports in the media and other real-life situations and identify potential sources of bias\ <u>Success Criteria</u> Univariate Data Analysis • Represent and analyse boxplots, explaining differences between multiple datasets in terms of shape, spread, and centre. • Compare or match the shapes of boxplots to distributions depicting the same data. • Determine the mean and standard deviation of a dataset (EXTENSION). • Investigate, analyse, and interpret the effect of individual data values, including outliers, on the standard deviation. (EXTENSION).		
7	<u>Content Descriptors</u> Analyse bivariate data represented in a two-way table, using proportions and comment on possible association between categorical variables <u>Success Criteria</u> Bivariate Data Analysis • Analyse bivariate data represented in a two-way table using proportions to comment on possible associations between categorical variables. • Represent the relationship between bivariate data in a scatter plot. • Draw a trend line by eye where appropriate to model relationships.	9F – 9I	

	- Use the graph and context to describe associations in terms of strength, direction, linearity, and outliers. - Make predictions from the model and explain any limitations (interpolation vs. extrapolation).		
8	<u>Content Descriptors</u> Represent the relationship between bivariate data in a scatter plot and draw a trend line by eye if appropriate. Use the graph and context to describe any association in terms of strength, direction, linearity and outliers. Make predictions and recognise and explain any limitations of the model <u>Success Criteria</u> Bivariate Data Analysis - Analyse bivariate data represented in a two-way table using proportions to comment on possible associations between categorical variables. - Represent the relationship between bivariate data in a scatter plot. - Draw a trend line by eye where appropriate to model relationships. - Use the graph and context to describe associations in terms of strength, direction, linearity, and outliers. - Make predictions from the model and explain any limitations (interpolation vs. extrapolation).	9F – 9I	
9	<u>Content Descriptors</u> Critically analyze the claims, inferences and conclusions of statistical reports in the media and other real-life situations and identify potential sources of bias <u>Success Criteria</u> Critical Statistical Analysis	9F – 9I	Week 9 Assessment 2 – Univariate and Bivariate Data (INVESTIGATION) Returned by Term 2 Week 2

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- Critically analyse the claims, inferences, and conclusions of statistical reports in the media and other real-life situations.
- Identify potential sources of bias in statistical reporting.

Assessment 2 – Univariate and Bivariate Data (INVESTIGATION)

TERM TWO			
Week	Content	Cambridge ESSENTIALS	Assessment
TRIGONOMETRY AND PYTHAGORAS			
1	<u>Content Descriptors</u> Use Pythagoras' theorem and/or trigonometry to determine unknown sides and angles in right-angled triangles involving angles of elevation and depression <u>Success Criteria</u> Right-Angled Trigonometry (2D) - Use Pythagoras' theorem and/or trigonometry to determine unknown sides and angles in right-angled triangles involving angles of elevation and depression. - Apply right-angled trigonometry to two-dimensional situations involving navigational bearings.	4F, 6A – 6E	
2	<u>Content Descriptors</u> Apply right-angled trigonometry to two dimensional situations involving navigational bearings <u>Success Criteria</u> 3D Applications Explore and apply Pythagoras' theorem and trigonometry to simple situations involving right-angled triangles in three-dimensional contexts projected to two dimensions. (EXTENSION)	4F, 6A – 6E	

3	<u>Content Descriptors</u> Explore and apply Pythagoras' theorem and trigonometry to simple situations involving right-angled triangles in three-dimensional contexts projected to two-dimensions (Year 9 Extension) <u>Success Criteria</u> 3D Applications Explore and apply Pythagoras' theorem and trigonometry to simple situations involving right-angled triangles in three-dimensional contexts projected to two dimensions. (EXTENSION)	4F, 6A – 6E	
4	Exam Revision Review Linear Functions, Univariate/Bivariate Data, Pythagoras and Trigonometry		
5	Assessment 3 - Semester 1 Exam		Assessment 3 - Semester 1 Exam Returned by Week 7
MEASUREMENT			
6	<u>Content Descriptors</u> Investigate, explore and determine the effect on the perimeter and area of shapes when they are enlarged or reduced by a scale factor <u>Success Criteria</u>	4E, 4G, 4H – 4N	

	Similarity and Scaling - Investigate and determine the effect on the perimeter and area of shapes when they are enlarged or reduced by a scale factor. - Investigate to determine the effect on surface area and volume when objects are enlarged or reduced by a scale factor.		
7	<u>Content Descriptors</u> Explore, explain and apply efficient strategies and formulas to determine the surface area and volume of right pyramids, right cones, spheres and related composite solids\ Use efficient strategies and apply formulas to determine the volume, capacity and surface area of composite solids, using appropriate units Investigate, calculate and identify the impact of errors on the accuracy and outcome of results in measurement situations <u>Success Criteria</u> Surface Area and Volume - Use efficient strategies and apply formulas to determine the volume, capacity, and surface area of composite solids. - Explore, explain, and apply efficient strategies and formulas to determine the surface area and volume of right pyramids, right cones, spheres, and related composite solids. (EXTENSION) Accuracy and Error	4E, 4G, 4H – 4N	

	Investigate, calculate, and identify the impact of errors on the accuracy and outcome of results in measurement situations		
8	<u>Content Descriptors (cont.)</u> Explore, explain and apply efficient strategies and formulas to determine the surface area and volume of right pyramids, right cones, spheres and related composite solids\ Use efficient strategies and apply formulas to determine the volume, capacity and surface area of composite solids, using appropriate units Investigate, calculate and identify the impact of errors on the accuracy and outcome of results in measurement situations <u>Success Criteria (cont.)</u> Surface Area and Volume - Use efficient strategies and apply formulas to determine the volume, capacity, and surface area of composite solids. - Explore, explain, and apply efficient strategies and formulas to determine the surface area and volume of right pyramids, right cones, spheres, and related composite solids. (EXTENSION) Accuracy and Error	4E, 4G, 4H – 4N	

	- Investigate, calculate, and identify the impact of errors on the accuracy and outcome of results in measurement situations Modelling with measurement and geometry In real-world situations involving Pythagoras' theorem, trigonometry and angles of elevation and depression in right-angled triangles, volume, capacity and surface area of composite objects, the impact of errors in measurement and/or the effect of enlargement and reduction on perimeter, area and volume of similar figures and objects - analyse the situation, decide if an exact or approximate solution is required and determine assumptions and constraints - represent the situation mathematically in order to reach a solution - interpret and communicate findings in terms of the context and any assumptions or constraints		
PROBABILITY			
9	<u>Content Descriptors</u> Set Notation Choose and construct appropriate sample spaces to show outcomes for two- and three-stage chance experiments both with and without replacement. Assign probabilities to events involving conditional statements, such as 'if ... then', 'given', 'of', 'knowing that' <u>Success Criteria</u>	8A – 8G	

	Sample Spaces and Events - Choose and construct appropriate sample spaces to show outcomes for two- and three-stage chance experiments both with and without replacement. Set Notation and Venn Diagrams - Use set notation to describe events, including the intersection ($A \cap B$), union ($A \cup B$), and complement (A') of events. (EXTENSION) - Represent events and calculate probabilities using Venn diagrams and two-way tables. - Identify and define mutually exclusive events (where the probability of $A \cap B$ is zero). (EXTENSION) - Apply the addition rule for probability: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. (EXTENSION)		
10	<u>Content Descriptors</u> Use weighted tree diagrams and/or formulas to assign probabilities to two- and three-stage chance events including situations involving conditional probability Conduct repeated chance experiments and simulations to model conditional probability and produce datasets using digital tools. Discuss, compare and analyse variation and estimated probabilities for conditional events Represent secondary data in two-way tables or Venn diagrams and assign probabilities to outcomes involving conditional statements <u>Success Criteria</u>	8A – 8G	

	Conditional Probability - Assign probabilities to events involving conditional statements, such as 'if ... then', 'given', 'of', 'knowing that'. - Use weighted tree diagrams and/or formulas to assign probabilities to two- and three-stage chance events, including situations involving conditional probability. Modelling with probability and statistics In real-world situations involving two- and three-stage chance experiments both with and without replacement, conditional probability or statements, boxplots, bivariate data and/or two-way tables - analyse the situation, pose questions as required and determine assumptions and constraints - determine appropriate production of a valid and reliable dataset, statistical measures, data representations and analyses, including examination of distributions, to effectively investigate the situation - interpret, draw inferences and communicate findings, in terms of the context, assumptions, constraints, chance variation and knowledge or insights gained		
11	<u>Content Descriptors</u> Assessment 4 – Probability (INVESTIGATION)		Week 11 Assessment 4 – Probability (INVESTIGATION) Returned by Term 3 Week 2

TERM THREE			
Week	Content	Cambridge ESSENTIALS 10/10A	Assessment
Algebra			
1	<u>Content Descriptors</u> Extend and apply index laws with positive-integer indices and variable bases, to include negative-integer indices Simplify algebraic products and quotients involving indices with integer and fractional indices Establish the connection between fractional indices and surds. Perform the four operations with surds and rationalise the denominator if required <u>Success Criteria</u> Indices and Surds - Extend and apply index laws with positive-integer indices and variable bases to include negative-integer indices. (EXTENSION) - Simplify algebraic products and quotients involving indices with integer and fractional indices. (EXTENSION) - Establish the connection between fractional indices and surds. (EXTENSION) - Perform the four operations with surds and rationalise the denominator if required. (EXTENSION)	3A – 3D, 4A – 4D, 5A – 5F	

2	<u>Content Descriptors</u> Explore efficient strategies to simplify expressions that involve addition, subtraction, multiplication or division of algebraic fractions with an algebraic expression in the numerator and/or denominator, including the use of factorisation <u>Success Criteria</u> Algebraic Fractions - Simplify algebraic fractions with an algebraic expression in the numerator and/or denominator, including the use of factorisation. (EXTENSION) - Explore efficient strategies to simplify expressions that involve addition, subtraction, multiplication, or division of algebraic fractions. (EXTENSION)	3A – 3D, 4A – 4D, 5A – 5F	
3	<u>Content Descriptors</u> Factorise monic and non-monic quadratic expressions using techniques, such as completing the square, perfect squares, difference of squares and grouping in pairs for four-term expressions <u>Success Criteria</u> Factorisation - Factorise monic quadratic expressions using techniques such as difference of squares and perfect squares. - Factorise non-monic quadratic expressions using techniques such as grouping in pairs. (EXTENSION)	3A – 3D, 4A – 4D, 5A – 5F	

	Factorise quadratic expressions by completing the square. (EXTENSION)		
4	<u>Content Descriptors</u> Assessment 5 – Algebra Skills (TEST)		Week 4 - Algebra Skills (TEST) Returned by Week 6
FUNCTIONS AND RELATIONS			
5	<u>Content Descriptors</u> Substitute values into real-life linear, quadratic or simple exponential formulas to find unknowns using digital tools Extend and apply knowledge of the expansion of binomial products to explore the factorisation of monic quadratics Solve monic and non-monic quadratic equations graphically and algebraically, including the use of the quadratic formula, factorising techniques and digital tools and verify the solution/s by substitution <u>Success Criteria</u> Quadratic Equations - Solve monic quadratic equations graphically and algebraically using factorising techniques. - Solve non-monic quadratic equations graphically and algebraically. (EXTENSION) - Solve quadratic equations using the quadratic formula. (EXTENSION)	5G – 5J, 3E – 3G, 7A – 7H, 10A – 10C, 10I – 10J	

6	<u>Content Descriptors (cont.)</u> Substitute values into real-life linear, quadratic or simple exponential formulas to find unknowns using digital tools Extend and apply knowledge of the expansion of binomial products to explore the factorisation of monic quadratics Solve monic and non-monic quadratic equations graphically and algebraically, including the use of the quadratic formula, factorising techniques and digital tools and verify the solution/s by substitution <u>Success Criteria (cont.)</u> Quadratic Equations - Solve monic quadratic equations graphically and algebraically using factorising techniques. - Solve non-monic quadratic equations graphically and algebraically. (EXTENSION) Solve quadratic equations using the quadratic formula. (EXTENSION)	5G – 5J, 3E – 3G, 7A – 7H, 10A – 10C, 10I – 10J	
7	<u>Content Descriptors</u> Use a table of values to plot points and graph quadratic functions of the form $y = ax^2 + c$. Identify and relate key graphical and algebraic features and make connections to the graphical and algebraic solution/s of $ax^2 + c = k$. Use digital tools to explore the shapes, features and related solutions to more complex quadratic functions Explore and use strategies, including digital tools, to model the equation of a quadratic function from a table of values or graph	5G – 5J, 3E – 3G, 7A – 7H, 10A – 10C, 10I – 10J	

	Graph monic and non-monic quadratics of the form $y = ax^2 + bx + c$ or $y = a(x - p)^2 + q$ and their transformations, manually and using digital tools. Identify and connect key graphical and algebraic features and make connections to the algebraic solution/s <u>Success Criteria</u> Graphs of Functions Quadratics: Graph monic quadratics of the form $y = ax^2 + bx + c$ or $y = a(x - p)^2 + q$ and their transformations. (EXTENSION)		
8	<u>Content Descriptors</u> Use a table of values to plot points and graph exponential functions of the form $y = a^x$ where $a > 0$. Identify and relate key graphical and algebraic features and use these to determine graphical solutions of related equations. Use digital tools to explore the shapes, features and related solutions to more complex exponential functions Identify and distinguish between linear, quadratic and exponential functions represented by equations, tables of values and graphs Use algebraic techniques to solve exponential equations that involve terms with related bases <u>Success Criteria</u> Graphs of Functions Exponentials: Use a table of values to plot points and graph exponential functions of the form $y = a^x$ where $a > 0$. (EXTENSION)	5G – 5J, 3E – 3G, 7A – 7H, 10A – 10C, 10I – 10J	

	Exponential Equations Use algebraic techniques to solve exponential equations that involve terms with related bases. (EXTENSION)		
9	<u>Content Descriptors</u> Explore, describe and interpret circles and rectangular hyperbolas where the asymptotes are parallel to the axes, and their transformations, using digital tools and make connections between the algebraic and graphical representations <u>Success Criteria</u> Graphs of Functions Circles and Hyperbolas: Explore, describe, and interpret circles and rectangular hyperbolas where the asymptotes are parallel to the axes. (EXTENSION)	5G – 5J, 3E – 3G, 7A – 7H, 10A – 10C, 10I – 10J	
10	<u>Content Descriptors</u> Revise Contents from Week 5 – 9 Assessment 6 – Functions and Relations (TEST)		Week 10 Assessment 6 – Functions and Relations (TEST) Returned by Term 4 Week 2

TERM FOUR			
Week	Content	Cambridge ESSENTIALS 10/10A	Assessment
NON-RIGHT TRIGONOMETRY			
1	<u>Content Descriptors</u> Explore to establish and use the sine, cosine and area rule to determine unknown sides and angles for any triangle <u>Success Criteria</u> Non-Right Triangle Trigonometry Explore to establish and use the sine, cosine, and area rules to determine unknown sides and angles for any triangle. (EXTENSION)	6F – 6K	
2	<u>Content Descriptors (cont.)</u> Explore to establish and use the sine, cosine and area rule to determine unknown sides and angles for any triangle <u>Success Criteria (cont.)</u> Non-Right Triangle Trigonometry Explore to establish and use the sine, cosine, and area rules to determine unknown sides and angles for any triangle. (EXTENSION)	6F – 6K	

3	<u>Content Descriptors</u> Use the unit circle and dynamic geometry software to explore and represent trigonometric functions graphically <u>Success Criteria</u> The Unit Circle and Functions - Use the unit circle and dynamic geometry software to explore and represent trigonometric functions graphically. (EXTENSION) - Solve simple trigonometric equations graphically, algebraically, or using the unit circle. (EXTENSION)	6F – 6K	
4	<u>Content Descriptors (cont.)</u> Use the unit circle and dynamic geometry software to explore and represent trigonometric functions graphically <u>Success Criteria (cont.)</u> The Unit Circle and Functions Use the unit circle and dynamic geometry software to explore and represent trigonometric functions graphically. (EXTENSION) Solve simple trigonometric equations graphically, algebraically, or using the unit circle. (EXTENSION)	6F – 6K	
5	Semester 2 Exam Revision (Includes content covered after Semester 1 Exams)		

6	Assessment 7 – Semester 2 Exams		Assessment 7 – Semester 2 Exams Returned by Week 7
7	Semester 2 Exam Review		
8 – 9	Year 11 Rollover		
10	Financial Mathematics Explore, explain and calculate income tax, including the use of tax tables Apply repeated simple interest to develop the compound interest formula and solve problems that relate to saving and borrowing <u>Success Criteria</u> Income Tax - Explore, explain, and calculate income tax, including the use of tax tables. Compound Interest - Apply repeated simple interest to develop the compound interest formula and solve problems that relate to saving and borrowing.	3H – 3J	

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ASSESSMENT OUTLINE

Task	Assessment	Type	Term/Week
1	Assessment 1 – Linear Functions	Test	T1W5
2	Assessment 2 – Univariate and Bivariate Data	Investigation	T1W9
3	Assessment 3 - Semester 1 Exam	Exam	T2W5
4	Assessment 4 – Probability	Investigation	T2W11
5	Assessment 5 – Algebra Skills	Test	T3W4
6	Assessment 6 – Functions and Relations	Test	T3W10
7	Semester 2 Exams	Exam	T4W6